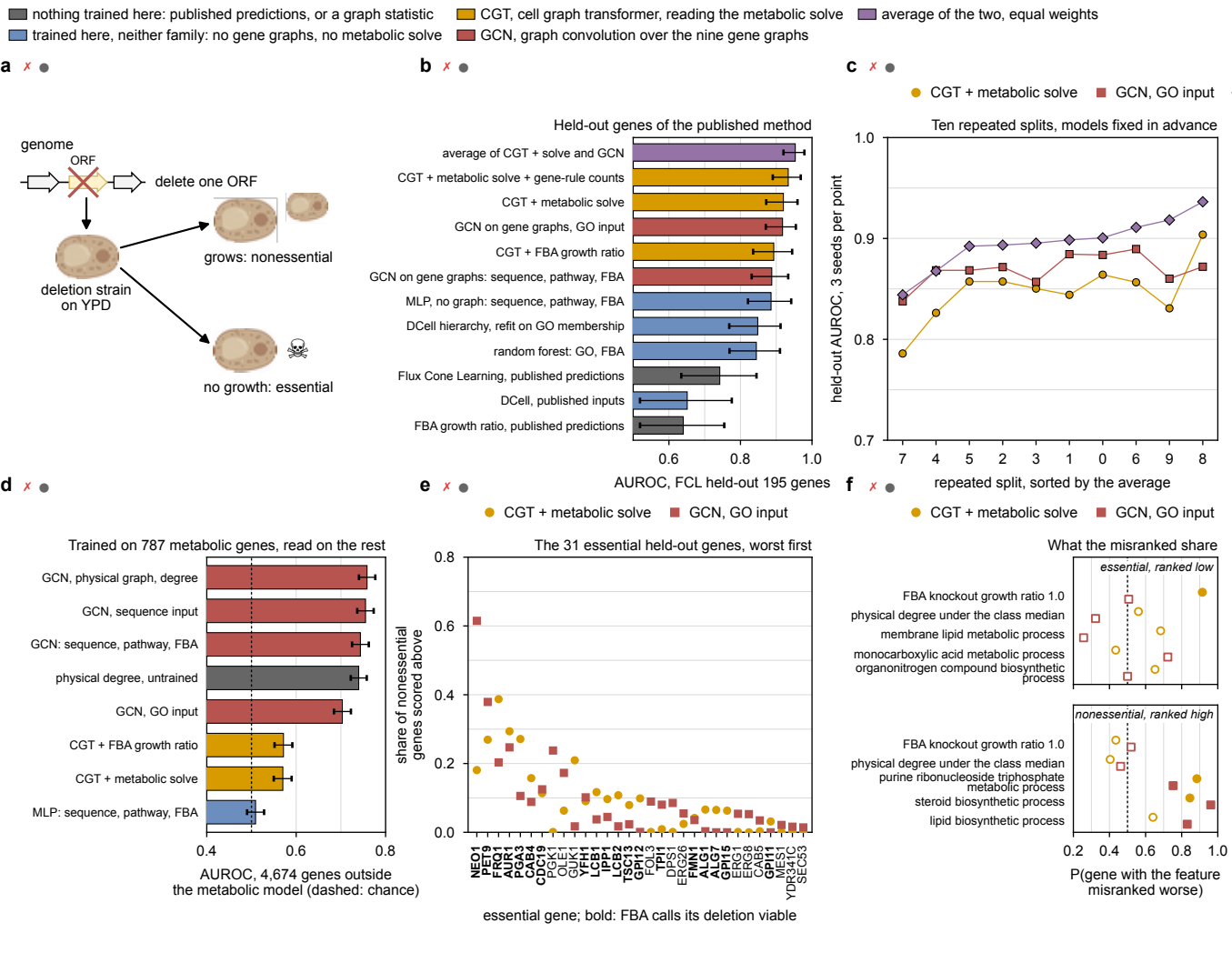

nothing trained here: published predictions, or a graph statistic
trained here, neither family: no gene graphs, no metabolic solve
CGT, cell graph transformer, reading the metabolic solve
GCN, graph convolution over the nine gene graphs
average of the two, equal weights
a
genome
ORF
delete one ORF
grows: nonessential
deletion strain on YPD
no growth: essential
b
Held-out genes of the published method
average of CGT + solve and GCN
CGT + metabolic solve + gene-rule counts
CGT + metabolic solve
GCN on gene graphs, GO input
CGT + FBA growth ratio
GCN on gene graphs: sequence, pathway, FBA
MLP, no graph: sequence, pathway, FBA
DCell hierarchy, refit on GO membership
random forest: GO, FBA
Flux Cone Learning, published predictions
DCell, published inputs
FBA growth ratio, published predictions
0.6
0.8
1.0
AUROC, FCL held-out 195 genes
c
CGT + metabolic solve
GCN, GO input
Ten repeated splits, models fixed in advance
held-out AUROC, 3 seeds per point
1.0
0.9
0.8
0.7
7 4 5 2 3 1 0 6 9 8
repeated split, sorted by the average
d
Trained on 787 metabolic genes, read on the rest
GCN, physical graph, degree
GCN, sequence input
GCN: sequence, pathway, FBA
physical degree, untrained
GCN, GO input
CGT + FBA growth ratio
CGT + metabolic solve
MLP: sequence, pathway, FBA
0.4
0.6
0.8
AUROC, 4,674 genes outside the metabolic model (dashed: chance)
e
CGT + metabolic solve
GCN, GO input
The 31 essential held-out genes, worst first
share of nonessential genes scored above
0.8
0.6
0.4
0.2
0.0
NEO1 PET9 FRQ1 ANP1 PGA3 CAB4 CDC19 PGK1 OLE1 GPI1 YPH1 LCB1 IPP1 LCB2 TSC13 GPI12 TPI1 DPS1 ERG26 FMN1 ALG1 ALG7 GPI15 GPI2 ERG8 CAB5 GPI11 MES1 YDR341C SEC53
essential gene; bold: FBA calls its deletion viable
f
CGT + metabolic solve
GCN, GO input
What the misranked share
essential, ranked low
FBA knockout growth ratio 1.0
physical degree under the class median
membrane lipid metabolic process
monocarboxylic acid metabolic process
organonitrogen compound biosynthetic process
nonessential, ranked high
FBA knockout growth ratio 1.0
physical degree under the class median
purine ribonucleoside triphosphate metabolic process
steroid biosynthetic process
lipid biosynthetic process
0.2
0.4
0.6
0.8
1.0
P(gene with the feature misranked worse)